1 Constitutive Foundation

The *Constitutive Foundation* provides the foundational semantic structure based on exact truthmaker semantics. Evaluation is hyperintensional, distinguishing propositions that agree on truth-value across all possible worlds but differ in their exact verification and falsification conditions.

1.1 Syntactic Primitives

The Constitutive Foundation interprets the following syntactic primitives:

- **Variables**: $x, y, z, \ldots$ (ranging over states)
- **Individual constants**: $a, b, c, \ldots$ (0-place function symbols)
- **n-place function symbols**: $f, g, h, \ldots$
- **n-place predicates**: $F, G, H, \ldots$
- **Sentence letters**: $p, q, r, \ldots$ (0-place predicates)
- **Lambda abstraction**: $\lambda x.\varphi$ (binding variable x in formula φ)
- **First-order identity**: $=$ (identity between terms)
- **Logical connectives**: $\neg, \wedge, \vee, \top, \bot, \equiv$

Definition 1.1 (Well-Formed Formulas)**.** The set of *well-formed formulas* is defined inductively:

$$\begin{aligned}\varphi ::= {} & p \mid F(t_1, \ldots, t_n) \mid \top \mid \bot \\ & \mid \neg\varphi \mid (\varphi \wedge \psi) \mid (\varphi \vee \psi) \\ & \mid (\varphi \equiv \psi) \mid (\lambda x.\varphi)(t)\end{aligned}$$

where p ranges over sentence letters, F over n-place predicates, t, t_i over terms, and x over variables.

See `Logos.SubTheories.Foundation.Syntax.ConstitutiveFormula` for the Lean implementation.

1.2 Derived Operators

Definition 1.2 (Material Conditional)**.**

$$\varphi \to \psi := \neg\varphi \vee \psi$$

Definition 1.3 (Biconditional)**.**

$$\varphi \leftrightarrow \psi := (\varphi \to \psi) \wedge (\psi \to \varphi)$$

Remark 1.4 (Quantifier Notation). First-order quantifiers are *not* primitive in the Constitutive Foundation. Generalized quantification is expressed via lambda abstraction:

- Universal: $\forall x.\varphi$ abbreviates $\forall(\lambda x.\varphi)$ where $\forall$ is a second-order predicate on properties
- Existential: $\exists x.\varphi$ abbreviates $\exists(\lambda x.\varphi)$ where $\exists$ is a second-order predicate on properties

First-order quantification is introduced in the Dynamical Foundation layer.

See `Logos.Dynamics.Syntax.CoreFormula` for the Lean implementation.

1.3 Constitutive Frame

Definition 1.5. A *constitutive frame* is a structure $\mathcal{F} = \langle \mathcal{S}, \sqsubseteq \rangle$ where:

- $\mathcal{S}$ is a nonempty set of *states*
- $\sqsubseteq$ is a partial order on $\mathcal{S}$ where $\langle \mathcal{S}, \sqsubseteq \rangle$ is a complete lattice

Remark 1.6. The constitutive frame is non-modal: neither possibility nor compatibility can be defined since this requires the task relation introduced in the Dynamical Foundation. Taking S itself to consist of *possible* states disrupts closure under fusion makes the conjunction of incompatible propositions empty, stripping the semantics of its capacity to propositions differing in subject-matter with no gain in simplicity or elegance.

Remark 1.7. The lattice structure provides:

- **Null state** $\square$: The bottom element (fusion of the empty set)
- **Full state** $\blacksquare$: The top element (fusion of all states)
- **State Fusion** $s \sqcup t$: The least upper bound of s and t
- **Pointwise Fusion** $f \sqcup g(s_1, \ldots, s_n) := f(s_1, \ldots, s_n) \sqcup g(s_1, \ldots, s_n)$

See `Logos.Foundation.Frame.ConstitutiveFrame` for the implementation.

1.4 Constitutive Model

Definition 1.8. A *constitutive model* is a structure $\mathcal{M} = \langle \mathcal{S}, \sqsubseteq, |\cdot| \rangle$ where $\langle \mathcal{S}, \sqsubseteq \rangle$ is a constitutive frame and $|\cdot|$ is an *interpretation function* satisfying:

- **n-place function symbols** $|f| : (\texttt{Fin}\ n \to \mathcal{S}) \to \mathcal{S}$
- **n-place predicates** $|F|^+, |F|^- : ((\texttt{Fin}\ n \to \mathcal{S}) \to \mathcal{S}) \to \mathsf{Prop}$ are the *verifier function* and *falsifier function* types which are required to satisfy:[1]

[1] The Lean implementation uses the mathematically equivalent Set-based formulation

- $f \sqcup g : |F|^{\pm}$ for any $f, g : |F|^{\pm}$.
- $\bigsqcup(a_1, \ldots, a_n) \sqsubseteq f(a_1, \ldots, a_n)$ for any $f : |F|^{\pm}$ and $a_1, \ldots, a_n : \mathcal{S}$.

Remark 1.9. Constants and sentence letters are covered by the trivial cases of 0-place functions and 0-place predicates:

- **Constant symbols** $|c| : \mathcal{S}$ is the type for 0-place functions.
- **Sentence letters** $|p|^+, |p|^- : \mathcal{S} \to \mathsf{Prop}$ are the *verifier function type* and *falsifier function type* for 0-place predicates.

See `Logos.Foundation.Model.ConstitutiveModel` for the implementation.

1.5 Variable Assignment

Definition 1.10. A *variable assignment* $\sigma : \mathrm{Var} \to \mathcal{S}$ maps each variable name to a state in the state space.

Notation 1.11. Greek letters ($\tau, \alpha, \beta, \ldots$) are reserved for world-histories. The letter σ (with subscripts $\sigma_1, \sigma_2, \ldots$) denotes variable assignments, chosen for compatibility across LaTeX, markdown, and Lean notation.

Definition 1.12 (Term Algebra)**.** The set Term of *terms* is defined inductively:

$$t ::= x \mid c \mid f(t_1, \ldots, t_n)$$

where x ranges over variables, c over individual constants (0-place function symbols), and f over n-place function symbols.

The *free variables* $\mathsf{FV}(t)$ of a term are defined by:

$$\begin{aligned} \mathsf{FV}(x) &= \{x\} \\ \mathsf{FV}(c) &= \varnothing \\ \mathsf{FV}(f(t_1, \ldots, t_n)) &= \bigcup_{i=1}^{n} \mathsf{FV}(t_i) \end{aligned}$$

Substitution $t[s/x]$ (replacing x with s in t) is defined by:

$$\begin{aligned} y[s/x] &= \begin{cases} s & \text{if } y = x \\ y & \text{otherwise} \end{cases} \\ c[s/x] &= c \\ f(t_1, \ldots, t_n)[s/x] &= f(t_1[s/x], \ldots, t_n[s/x]) \end{aligned}$$

`Set` $((\mathtt{Fin}\ n \to \mathcal{S}) \to \mathcal{S})$ since in Lean 4, `Set` α is defined as $\alpha \to \mathsf{Prop}$. The Set notation emphasizes the collectional aspect while the Prop notation emphasizes the logical aspect; both capture the same mathematical structure.

See `Logos.SubTheories.Foundation.Syntax.Term` for the Lean implementation.

Definition 1.13. The *x-variant* $\sigma[v/x]$ of the variable assignment σ differs at most from σ by assigning x to v.

Definition 1.14. The *extension* of a term given a model $\mathcal{M}$ and assignment σ is a function $[\![\cdot]\!]^{\sigma}_{\mathcal{M}} : \mathsf{Term} \to \mathcal{S}$ defined by:

- **Variable** $[\![x]\!]^{\sigma}_{\mathcal{M}} = \sigma(x)$
- **Function application** $[\![f(t_1, \ldots, t_n)]\!]^{\sigma}_{\mathcal{M}} = |f|(\lambda i : \texttt{Fin}\ n, [\![t_i]\!]^{\sigma}_{\mathcal{M}})$

See `Logos.Foundation.Assignment.Assignment` for the implementation.

1.6 Verification and Falsification

The *verification* ($\Vdash$) and *falsification* ($\dashv\!\Vert$) of a formula φ is defined recursively relative to a model $\mathcal{M}$, assignment σ, and state s.

1.6.1 Atomic Formulas

Definition 1.15 (Atomic Verification and Falsification)**.**

$$\mathcal{M}, \sigma, s \Vdash F(t_1, \ldots, t_n) \ \textit{iff}\ \exists f \in |F|^{+} : s = f(\lambda i : \texttt{Fin}\ n, [\![t_i]\!]^{\sigma}_{\mathcal{M}}) \tag{1}$$

$$\mathcal{M}, \sigma, s \dashv\!\Vert F(t_1, \ldots, t_n) \ \textit{iff}\ \exists f \in |F|^{-} : s = f(\lambda i : \texttt{Fin}\ n, [\![t_i]\!]^{\sigma}_{\mathcal{M}}) \tag{2}$$

1.6.2 Lambda Abstraction

Definition 1.16 (Lambda Verification and Falsification)**.**

$$\mathcal{M}, \sigma, s \Vdash (\lambda x.\varphi)(t) \ \textit{iff}\ \mathcal{M}, \sigma[[\![t]\!]^{\sigma}_{\mathcal{M}}/x], s \Vdash \varphi \tag{3}$$

$$\mathcal{M}, \sigma, s \dashv\!\Vert (\lambda x.\varphi)(t) \ \textit{iff}\ \mathcal{M}, \sigma[[\![t]\!]^{\sigma}_{\mathcal{M}}/x], s \dashv\!\Vert \varphi \tag{4}$$

1.6.3 Negation

Definition 1.17 (Negation Verification and Falsification)**.**

$$\mathcal{M}, \sigma, s \Vdash \neg\varphi \ \textit{iff}\ \mathcal{M}, \sigma, s \dashv\!\Vert \varphi \tag{5}$$

$$\mathcal{M}, \sigma, s \dashv\!\Vert \neg\varphi \ \textit{iff}\ \mathcal{M}, \sigma, s \Vdash \varphi \tag{6}$$

1.6.4 Conjunction

Definition 1.18 (Conjunction Verification and Falsification)**.**

$$\mathcal{M}, \sigma, s \Vdash \varphi \wedge \psi \ \textit{iff}\ s = t \sqcup u \text{ for some } t, u \text{ where } \mathcal{M}, \sigma, t \Vdash \varphi \text{ and } \mathcal{M}, \sigma, u \Vdash \psi \tag{7}$$

$$\mathcal{M}, \sigma, s \dashv \varphi \wedge \psi \ \textit{iff}\ \mathcal{M}, \sigma, s \dashv \varphi, \text{ or } \mathcal{M}, \sigma, s \dashv \psi, \text{ or } s = t \sqcup u \text{ for some } t, u \text{ where } \mathcal{M}, \sigma, t \dashv \varphi \text{ and } \mathcal{M}, \sigma, u \dashv \psi \tag{8}$$

1.6.5 Disjunction

Definition 1.19 (Disjunction Verification and Falsification)**.**

$$\mathcal{M}, \sigma, s \Vdash \varphi \vee \psi \ \textit{iff}\ \mathcal{M}, \sigma, s \Vdash \varphi, \text{ or } \mathcal{M}, \sigma, s \Vdash \psi, \text{ or } s = t \sqcup u \text{ for some } t, u \text{ where } \mathcal{M}, \sigma, t \Vdash \varphi \text{ and } \mathcal{M}, \sigma, u \Vdash \psi \tag{9}$$

$$\mathcal{M}, \sigma, s \dashv \varphi \vee \psi \ \textit{iff}\ s = t \sqcup u \text{ for some } t, u \text{ where } \mathcal{M}, \sigma, t \dashv \varphi \text{ and } \mathcal{M}, \sigma, u \dashv \psi \tag{10}$$

1.6.6 Top and Bottom

Definition 1.20 (Top and Bottom Verification and Falsification)**.**

$$\mathcal{M}, \sigma, s \Vdash \top \text{ for all } s \in \mathcal{S} \tag{11}$$

$$\mathcal{M}, \sigma, s \dashv \top \ \textit{iff}\ s = \blacksquare \ (\textit{full state}) \tag{12}$$

$$\mathcal{M}, \sigma, s \nVdash \bot \text{ for all } s \tag{13}$$

$$\mathcal{M}, \sigma, s \dashv \bot \ \textit{iff}\ s = \square \ (\textit{null state}) \tag{14}$$

1.6.7 Propositional Identity

Definition 1.21 (Propositional Identity Verification and Falsification)**.**

$$\begin{aligned} \mathcal{M}, \sigma, s \Vdash \varphi \equiv \psi \ \textit{iff}\ & s = \square \text{ and for all } t \text{ both:} \\ & (1)\ \mathcal{M}, \sigma, t \Vdash \varphi \text{ just in case } \mathcal{M}, \sigma, t \Vdash \psi; \text{ and} \\ & (2)\ \mathcal{M}, \sigma, t \dashv \varphi \text{ just in case } \mathcal{M}, \sigma, t \dashv \psi \end{aligned} \tag{15}$$

$$\begin{aligned} \mathcal{M}, \sigma, s \dashv \varphi \equiv \psi \ \textit{iff}\ & \text{at least one of the following:} \\ & (1)\ \mathcal{M}, \sigma, s \Vdash \varphi \text{ and } \mathcal{M}, \sigma, s \nVdash \psi; \\ & (2)\ \mathcal{M}, \sigma, s \nVdash \varphi \text{ and } \mathcal{M}, \sigma, s \Vdash \psi; \\ & (3)\ \mathcal{M}, \sigma, s \dashv \varphi \text{ and } \mathcal{M}, \sigma, s \not\dashv \psi; \text{ or} \\ & (4)\ \mathcal{M}, \sigma, s \not\dashv \varphi \text{ and } \mathcal{M}, \sigma, s \dashv \psi. \end{aligned} \tag{16}$$

See `Logos.Foundation.Semantics.verifies` for the implementation.

1.7 Essence and Ground

These constitutive relations can be defined via propositional identity:

Definition 1.22 (Essence and Ground)**.**

$$\varphi \sqsubseteq \psi := \varphi \wedge \psi \equiv \psi \qquad \text{“}\varphi \text{ is essential to } \psi\text{” (conjunctive part)} \tag{17}$$

$$\varphi \leqslant \psi := \varphi \vee \psi \equiv \psi \qquad \text{“}\varphi \text{ grounds } \psi\text{” (disjunctive part)} \tag{18}$$

Definition 1.23 (Reduction)**.** *Reduction* combines essence and ground:

$$\varphi \Rightarrow \psi := (\varphi \sqsubseteq \psi) \wedge (\varphi \leqslant \psi)$$

This expresses that φ is both a conjunctive part (essential to) and a disjunctive part (grounds) of ψ.

Remark 1.24. Essence and ground are interrelated by negation:

- $\varphi \sqsubseteq \psi$ *iff* $\neg\varphi \leqslant \neg\psi$
- $\varphi \leqslant \psi$ *iff* $\neg\varphi \sqsubseteq \neg\psi$

See `Logos.Foundation.Relations.essence` for the implementation.

1.8 Bilateral Propositions

Definition 1.25 (Bilateral Proposition)**.** A *bilateral proposition* P is an ordered pair $\langle V, F\rangle$ where V and F are subsets of $\mathcal{S}$ closed under fusion:

- **Fusion closure for verifiers**: $s \sqcup t \in V$ for any $s, t \in V$
- **Fusion closure for falsifiers**: $s \sqcup t \in F$ for any $s, t \in F$

Remark 1.26 (Bilattice Structure). The space of bilateral propositions forms a non-interlaced bilattice where:

- $\langle P, \sqsubseteq\rangle$ and $\langle P, \leqslant\rangle$ are complete lattices
- Negation exchanges the two orderings: $X \leqslant Y$ iff $\neg X \sqsubseteq \neg Y$
- Conjunction and disjunction are the least upper bounds with respect to $\sqsubseteq$ and $\leqslant$ respectively

Definition 1.27 (Propositional Operations)**.** For sets of states $X, Y \subseteq \mathcal{S}$ and bilateral propositions $P = \langle P^+, P^-\rangle$ and $Q = \langle Q^+, Q^-\rangle$, we may define the operations on their verifier and falsifier sets as follows:

- **Pairwise Fusion**: $X \otimes Y := \{s \sqcup t : s \in X, t \in Y\}$
- **Closed Fusion**: $X \oplus Y := X \cup Y \cup (X \otimes Y)$

- **Conjunction**: $P \wedge Q := \langle P^+ \otimes Q^+, P^- \oplus Q^- \rangle$
- **Disjunction**: $P \vee Q := \langle P^+ \oplus Q^+, P^- \otimes Q^- \rangle$
- **Negation**: $\neg P := \langle P^-, P^+ \rangle$

See `Logos.Foundation.Proposition.BilateralProp`.

1.9 Logical Consequence

Definition 1.28 (Logical Consequence)**.** The *Constitutive Foundation* of the Logos is restricted to propositional identity sentences of the form $\varphi \equiv \psi$ where φ and ψ are atomic sentences or built from $\neg, \wedge, \vee, \equiv$.

$$\begin{aligned} \Gamma \vDash \varphi \ \textit{iff} \ & \text{for any model } \mathcal{M} \text{ and assignment } \sigma : \\ & \text{if } \mathcal{M}, \sigma, \square \vDash \psi \text{ for all } \psi \in \Gamma, \text{ then } \mathcal{M}, \sigma, \square \vDash \varphi \end{aligned}$$

Remark 1.29 (Identity and Contingency). The *Constitutive Foundation* does not provide the semantic resources to evaluate non-identity sentences which depend on contingent states of affairs rather than purely structural relations in state space. The *Dynamical Foundation* encodes contingencies by including a world-history, time, and list of temporal references in addition to a variable assignment in the contexts at which sentences are evaluated.